Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

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Worked Solutions **Introduction to Logarithms and Their Properties**

Q001 Written

Convert the source forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Solution

1. Use $a^p = M \iff \log_a M = p$; swap the last two numbers.

Final answer

$\log_2 32 = 5;\ 3^2 = 9$

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Solution

1. Identify base, exponent, and argument; then interchange the exponent and argument positions.

Final answer

$\log_2 64 = 6; \log_4(1/16) = -2$, with the same swap rule for the remaining items.

Worked Solutions **Introduction to Logarithms and Their Properties**

Q003 **Written**

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Solution

1. Check $a > 0,\ a \neq 1,\ M > 0$, then write the equivalent form without changing the values.

Final answer

Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 **Written**

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

Solution

1. First find x from $5^x = 25$, then recognise $2 = (1/2)^{-1}$.

Final answer

$x = 2, so \log_{1/2} 2 = -1.$

Worked Solutions **Introduction to Logarithms and Their Properties**

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1, \log_5 5, \log_2 8, \log_2 16, \log_5(1/5), \log_7(49^{3/2})$.

Solution

1. Rewrite each argument as a power of its base; the exponent is the logarithm.

Final answer

Values: $0, 1, 3, 4, -1, 3/2$.

Q006 Written

Find the six Try values $\log_5 1, \log_3 3, \log_4 4, \log_3 27, \log_2(1/4), \log_5(1/125)$.

Solution

1. Use $\log_a 1 = 0$, $\log_a a = 1$, and the power definition.

Final answer

$0, 1, 1, 3, -2, -3$.

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Q007 Written

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

Solution

1. Do not approximate. For reciprocals, use a negative exponent; for roots, use a fractional exponent.

Final answer

Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

Q008 Written

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

Solution

1. Since $a^2 = (a^3)^{2/3}$, the logarithm is 2/3.

Final answer

2/3.

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Q009 Written

Perform $\log_6 2 + \log_6 3$ and $\log_2 30 - \log_2 3 - \log_2 5$.

Solution

1. Combine products for addition and quotients for subtraction, then express the result as a base power.

Final answer

1 and 1.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.

Solution

1. Move coefficients into exponents first, combine logs, simplify the argument, and read the final exponent.

Final answer

Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.

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Q011 Written

Perform all 18 source expressions (a)+(r), combining same-base logarithms exactly.

Solution

1. Apply the power law before product/quotient combination; simplify the final argument as a power of the base.

Final answer

Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3\log_5 \sqrt{2} - 2\log_5 \sqrt{6}$ to one reduced fraction.

Solution

1. Move coefficients inside, combine the argument, and reduce it to a power of 5.

Final answer

1.

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Q013 Written

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24, \log_{10}(27/4), \log_{10} 45, \log_{10}(24/75)$ in terms of a, b.

Solution

1. Prime-factorise the argument, use $5 = 10/2$ when required, then apply product, quotient, and power laws.

Final answer

$3a + b,\ 2 - 2a + 3b,\ 2a + 1 - a,\ b + 2 - 2a.$

Q014 Written

Given $\log_{10} 2 = a, \log_{10} 3 = b$, express the four Try arguments $72, 3^8/10, 15, 12$ in terms of a, b.

Solution

1. Write every integer as $2^m 3^n 5^p$, replace $\log_{10} 2, \log_{10} 3$, and convert the 5-factor using $1 - a$.

Final answer

Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

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Q015 **Written**

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Solution

1. Keep numerator and denominator factors separate until the final subtraction.

Final answer

Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 **Written**

If $\log_c 3 = x, \log_c 2 = y$, write $\log_c 12, \log_c(3/8), \log_c \sqrt{6}, \log_2 3, \log_{3b}(2^b c)$ in terms of x, y.

Solution

1. Factorise arguments, use the logarithm laws, and use change of base for the last two forms.

Final answer

$x + 2y,\ x - 3y,\ (x + y)/2,\ x/y,\ (1 + by)/(bx).$

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Q017 **Written**

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Solution

1. Use $\log_a b = \log_c b / \log_c a$, choosing a base that turns both numbers into powers.

Final answer

$3/2$ and -3.

Q018 **Written**

Evaluate the five Try logarithms using a suitable common base.

Solution

1. Choose the smallest convenient base; do not use decimal approximations when exact powers are available.

Final answer

Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

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Q019 **Written**

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Solution

1. Convert numerator and denominator to the same base, then cancel common factors.

Final answer

All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 **Written**

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.

Solution

1. Apply change of base to 2, then express the radical denominator as a power.

Final answer

$$\frac{\log_2 3 - 5/2}{2\log_2 3}.$$

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Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Solution

1. Make all logarithms use base 2 (or 3), then cancel adjacent change-of-base factors.

Final answer

Use reciprocal and power identities; the worked values are exact constants.

Q022 Written

Solve the four Try products of logarithms after making the bases the same.

Solution

1. Treat each logarithm as a change-of-base fraction and cancel before multiplying.

Final answer

Each product collapses to a rational number after change of base and cancellation.

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Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

Solution

1. Make the base 2, 3, or 5 consistently, simplify inside parentheses, then multiply.

Final answer

Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 Written

Simplify $\log_x(\sqrt{y})\ \log_{y^2}(z^3)\ \log_{\sqrt{z}}(x^4)$ for $x > 1$.

Solution

1. Convert each factor to a common natural-log ratio; exponents contribute $1/2, 3/2, 8$, whose product is 12.

Final answer

12.

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Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Solution

1. Use $\log_4(x^2) = 2\log_4|x|$; negative x belongs to f's domain but not g's.

Final answer

They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

Solution

1. Plot $y = -1, 0, 1$ in the exponential counterpart, then reflect in $y = x$.

Final answer

Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

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Q027 **Written**

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and 1/3.

Solution

1. Use the domain $x > 0$, vertical asymptote $x = 0$, and inverse reflection.

Final answer

Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 **Written**

Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.

Solution

1. Every graph passes through $(1, 0)$; do not draw it across the y-axis.

Final answer

Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

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Q029 **Written**

For $f(x) = \log_5(x-2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Solution

1. Translate the standard inverse pair by the same horizontal/vertical shifts and identify the resulting symmetry line.

Final answer

They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 **Written**

Order the displayed logarithmic sets, including $\log_2 3, \log_2 5, 2\log_2(1/2), \log_{1/2} 3, \log_{1/2} 5$.

Solution

1. Compare arguments only after the bases are the same and the base condition is stated.

Final answer

Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

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Q031 Written

Order the four Try sets of logarithms using monotonicity and change of base.

Solution

1. Rewrite $k \log_a M$ as $\log_a(M^k)$, then compare the arguments.

Final answer

Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 Written

Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.

Solution

1. Move coefficients inside first, then compare arguments with the correct direction.

Final answer

Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

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Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Solution

1. Express both logs with a common base and state the monotonicity case before writing the inequality.

Final answer

Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 MCQ

$\log_2 32$ equals:

Solution

1. Since $2^5 = 32$.

Correct option: C

Final answer

5

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Q035 MCQ

$\log_{1/2} 2$ **equals:**

Solution

1. $(1/2)^{-1} = 2$.

Correct option: B

Final answer

-1

Q036 MCQ

$\log_5(1/125)$ **equals:**

Solution

1. $1/125 = 5^{-3}$.

Correct option: A

Final answer

-3

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Q037 **MCQ**

$\log_7 49^{3/2}$ **equals:**

Solution

1. Write $49^{3/2} = 7^3$.

Correct option: C

Final answer

3/2

Q038 **MCQ**

$\log_a M + \log_a N$ **equals:**

Solution

1. Use the product law.

Correct option: B

Final answer

$\log_a(MN)$

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Q039 MCQ

The coefficient $3\log_5 2$ becomes:

Solution

1. Use $k\log_a M = \log_a(M^k)$.

Correct option: B

Final answer

$\log_5 2^3$

Q040 MCQ

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:

Solution

1. $24 = 2^3 \cdot 3$.

Correct option: C

Final answer

$3a + b$

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Q041 MCQ

$\log_c \sqrt{6}$ **in terms of** $x = \log_c 3, y = \log_c 2$ **is:**

Solution

1. Use $\sqrt{6} = 6^{1/2}$.

Correct option: B

Final answer

$(x + y)/2$

Q042 MCQ

$\log_{16} 64$ **equals:**

Solution

1. $16 = 2^4, 64 = 2^6$.

Correct option: B

Final answer

$3/2$

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Q043 MCQ

Change of base gives $\log_a b =$:

Solution

1. Use the quotient of common-base logs.

Correct option: B

Final answer

$\log_c b / \log_c a$

Q044 MCQ

$\log_4 16 \cdot \log_9 3$ equals:

Solution

1. Values are $2 \cdot 1/2$.

Correct option: A

Final answer

$1/2$

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Q045 MCQ

The challenge product of three logarithms equals:

Solution

1. Multiply exponent factors after change of base.

Correct option: C

Final answer

12

Q046 MCQ

The domain of $y = \log_a x$ is:

Solution

1. The argument must be positive.

Correct option: C

Final answer

$x > 0$

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Solution

1. Inverse graphs reflect in $y = x$.

Correct option: C

Final answer

$y = x$

Q048 MCQ

For $a > 1$ and $M < N$, which is true?

Solution

1. The logarithm is increasing.

Correct option: B

Final answer

$\log_a M < \log_a N$

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Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?

Solution

1. The logarithm is decreasing.

Correct option: A

Final answer

$\log_a M > \log_a N$

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:

Solution

1. Logarithmic graphs require $x > 0$.

Correct option: A

Final answer

$x = 0$

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Q051 MCQ

A negative coefficient $-2\log_a M$ can be written as:

Solution

1. Move the coefficient into the exponent.

Correct option: A

Final answer

$\log_a M^{-2}$

Q052 MCQ

The base of $\log_a M$ is:

Solution

1. The subscript is the base.

Correct option: B

Final answer

a

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Q053 MCQ

The argument of $\log_a M$ is:

Solution

1. The number inside the logarithm is the argument.

Correct option: B

Final answer

M

Q054 MCQ

$\log_2(1/8)$ **equals:**

Solution

1. $1/8 = 2^{-3}$.

Correct option: A

Final answer

-3

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Q055 MCQ

$\log_4 4$ **equals:**

Solution

1. The argument equals the base.

Correct option: B

Final answer

1

Q056 MCQ

$\log_3 1$ **equals:**

Solution

1. Every allowed base to power zero is 1.

Correct option: B

Final answer

0

Worked Solutions **Introduction to Logarithms and Their Properties**

Q057 MCQ

$\log_a M - \log_a N$ **equals:**

Solution

1. Use the quotient law.

Correct option: B

Final answer

$\log_a(M/N)$

Q058 MCQ

$2\log_3 5$ **equals:**

Solution

1. Move 2 into the argument as an exponent.

Correct option: B

Final answer

$\log_3 25$

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Q059 MCQ

If $a = \log_{10} 2, b = \log_{10} 3, \log_{10} 12$ is:

Solution

1. $12 = 2^2 \cdot 3.$

Correct option: B

Final answer

$2a + b$

Q060 MCQ

$\log_{10} 5$ **in terms of** $a = \log_{10} 2$ **is:**

Solution

1. $5 = 10/2.$

Correct option: B

Final answer

$1 - a$

Worked Solutions **Introduction to Logarithms and Their Properties**

Q061 MCQ

$\log_8 32$ **equals:**

Solution

1. $8 = 2^3, 32 = 2^5$.

Correct option: B

Final answer

5/3

Q062 MCQ

$\log_{25} 125$ **equals:**

Solution

1. $25 = 5^2, 125 = 5^3$.

Correct option: B

Final answer

3/2

Worked Solutions **Introduction to Logarithms and Their Properties**

Q063 MCQ

$\log_a b \cdot \log_b a$ **equals:**

Solution

1. Change of base makes reciprocal factors.

Correct option: B

Final answer

1

Q064 MCQ

$\log_2 8 \cdot \log_8 2$ **equals:**

Solution

1. The factors are reciprocals.

Correct option: B

Final answer

1

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Q065 *MCQ*

The best common base for 16 and 64 is:

Solution

1. Both numbers are powers of 2.

Correct option: A

Final answer

2

Q066 *MCQ*

The range of $y = \log_a x$ is:

Solution

1. Every real exponent is attained.

Correct option: B

Final answer

$\mathbb{R}$

Worked Solutions **Introduction to Logarithms and Their Properties**

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Solution

1. Swap the exponent and argument.

Correct option: A

Final answer

$\log_a M = p$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\log_2 8 + \log_2 4$ equals:

Solution

1. Combine to $\log_2 32$.

Correct option: C

Final answer

5

Worked Solutions **Introduction to Logarithms and Their Properties**

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ **equals:**

Solution

1. Use base 2 exponents 6/4.

Correct option: B

Final answer

3/2

Quiz MCQ 4 **Concept Quiz · MCQ**

For $a > 1, M < N$ **implies:**

Solution

1. The logarithmic function is increasing.

Correct option: B

Final answer

$\log_a M < \log_a N$

Worked Solutions **Introduction to Logarithms and Their Properties**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find $\log_3(1/27)$**.**

Solution

1. $1/27 = 3^{-3}$.

Final answer

-3

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\log_5 25 + \log_5 5$**.**

Solution

1. Combine to $\log_5 125 = 3$.

Final answer

3

Worked Solutions **Introduction to Logarithms and Their Properties**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Solution

1. $\log_2 64/\log_2 16 = 6/4 = 3/2$.

Final answer

3/2

Quiz WRITTEN 8 **Concept Quiz · Written**

State the domain and range of $y = \log_a x$.

Solution

1. The argument is positive and every real exponent is attained.

Final answer

Domain $x > 0$; range all real numbers

Answer key

Use the question number shown on each page.

Q001 $\log_2 32 = 5;\ 3^2 = 9$

Q002 $\log_2 64 = 6; \log_4(1/16) = -2$, with the same swap rule for the remaining items.

Q003 Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 $x = 2$, so $\log_{1/2} 2 = -1$.

Q005 Values: $0, 1, 3, 4, -1, 3/2$.

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Q007 Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

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Q012 1.

Q013 $3a + b,\ 2 - 2a + 3b,\ 2a + 1 - a,\ b + 2 - 2a$.

Q014 Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

Q015 Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 $x + 2y,\ x - 3y,\ (x + y)/2,\ x/y,\ (1 + by)/(bx)$.

Q017 $3/2$ and -3.

Q018 Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

Q019 All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 $\dfrac{\log_2 3 - 5/2}{2 \log_2 3}$.

Q021 Use reciprocal and power identities; the worked values are exact constants.

Q022 Each product collapses to a rational number after change of base and cancellation.

Q023 Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 12.

Q025 They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

Q027 Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

Q029 They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

Q031 Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

Q033 Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 C

Q035 B

Q036 A

Q037 C

Q038 B

Q039 B

Q040 C

Q041 B

Q042 B

Q043 B

Q044 A

Q045 C

Q046 C

Q047 C

Q048 B

Q049 A

Q050 A

Q051 A

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 B

Q058 B

Q059 B

Q060 B

Q061 B

Q062 B

Q063 B

Q064 B

Q065 A

Q066 B

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 -3

Quiz WRITTEN 6 3

Quiz WRITTEN 7 3/2

Quiz WRITTEN 8 Domain $x > 0$; range all real numbers